

3. Best practices for pipeline structure

Nextflow from zero to hero

Not only because of readability...

```
my_pipeline/  
├── modules/  
│   ├── fastp/  
│   │   └── main.nf  
│   └── bwa/  
│       └── main.nf  
├── workflows/  
│   ├── reads-qc/  
│   │   └── main.nf  
│   └── mapping/  
│       └── main.nf  
├── main.nf  
└── nextflow.config
```

Configuration

And the conf/ folder

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├── conf/
│   ├── base.config
│   └── modules.config
├── main.nf
└── nextflow.config
```

nextflow.config

- General configuration of the pipeline
 - parameters default (params)

```
params {
    // Input/output params
    input_file = '' //input file table TSV: sample_id, fastq_R1, fastq_R2
    reference_genome = '' //path to reference genome fasta file
    outdir = 'results' //output directory

    // Alignment QC params
    coverage_regions = null //bed file with regions to calculate coverage statistics (optional)

    // Variant calling params
    variant_caller = 'deepvariant' //variant caller to use: deepvariant, gatk-haplotypecaller
    variant_caller_use_gpu = true //whether to use GPU acceleration (if available)
    variant_mode = 'gvcf' //variant calling mode: vcf or gvcf
    save_gatk_realigned_bam = false //whether to save the GATK realigned BAM files
    merge_samples = true //whether to merge all samples into a single cohort VCF file
    deepvariant_model_type = 'WGS' //deepvariant model type: WGS, WES, PACBIO.
    variant_chromosomes = null //list of chromosomes to use in variant calling (if null all canonical chromosomes are used)
    variant_regions = null //bed file with regions to restrict variant calling to (optional)

    // Advanced params
    publish_mode = 'copy'

    // nf-core profiles config options
    custom_config_version = 'master'
    custom_config_base = "https://raw.githubusercontent.com/nf-core/configs/${params.custom_config_version}"
    hostnames = []
    config_profile_description = null
    config_profile_contact = null
    config_profile_url = null
    config_profile_name = null
}
```

nextflow.config

- General configuration of the pipeline
 - parameters default (params)
 - profiles (profiles)

```
// Computational environments
profiles {
  conda {
    conda.enabled      = true
    docker.enabled     = false
    singularity.enabled = false
    podman.enabled     = false
    shifter.enabled     = false
    charliecloud.enabled = false
    conda.channels     = ['conda-forge', 'bioconda']
    apptainer.enabled  = false
  }
  mamba {
    conda.enabled      = true
    conda.useMamba     = true
    docker.enabled     = false
    singularity.enabled = false
    podman.enabled     = false
    shifter.enabled     = false
    charliecloud.enabled = false
    apptainer.enabled  = false
  }
  docker {
    docker.enabled     = true
    conda.enabled      = false
    singularity.enabled = false
    podman.enabled     = false
    shifter.enabled     = false
    charliecloud.enabled = false
    apptainer.enabled  = false
    docker.runOptions  = '-u $(id -u):$(id -g)'
  }
  singularity {
    singularity.enabled = true
    singularity.autoMounts = true
    conda.enabled      = false
    docker.enabled     = false
    podman.enabled     = false
    shifter.enabled     = false
    charliecloud.enabled = false
    apptainer.enabled  = false
  }
  test {
    includeConfig 'conf/test.config'
  }
}
```

nextflow.config

- General configuration of the pipeline
 - parameters default (params)
 - profiles (profiles)
 - execution reports
 - timeline
 - report
 - trace
 - dag

```
// Set filename and location for timeline, report, trace and dag
def trace_timestamp = new java.util.Date().format( 'yyyy-MM-dd_HH-mm-ss' )
timeline {
    enabled = true
    file    = "${params.outdir}/pipeline_info/execution_timeline_${trace_timestamp}.html"
}
report {
    enabled = true
    file    = "${params.outdir}/pipeline_info/execution_report_${trace_timestamp}.html"
}
trace {
    enabled = true
    file    = "${params.outdir}/pipeline_info/execution_trace_${trace_timestamp}.txt"
}
dag {
    enabled = true
    file    = "${params.outdir}/pipeline_info/pipeline_dag_${trace_timestamp}.html"
}
```


nextflow.config

- General configuration of the pipeline
 - parameters default (params)
 - profiles (profiles)
 - execution reports
 - timeline
 - report
 - trace
 - dag
 - manifest (manifest)

```
manifest {  
  name           = 'nf-core/methylseq'  
  contributors   = [  
  ]  
  homepage       = 'https://github.com/nf-core/methylseq'  
  description    = ""Methylation (Bisulfite-Sequencing) Best Practice analysis pipeline, part of the nf-core community.""  
  mainScript     = 'main.nf'  
  defaultBranch  = 'master'  
  nextflowVersion = '!=25.04.0'  
  version        = '4.2.0'  
  doi            = '10.5281/zenodo.1343417'  
}
```

nextflow.config

- General configuration of the pipeline
 - parameters default (params)
 - profiles (profiles)
 - execution reports
 - timeline
 - report
 - trace
 - dag
 - manifest (manifest)
 - includeConfig

```
includeConfig 'conf/base.config'  
includeConfig 'conf/modules.config'
```


conf/base.config

- Included by *nextflow.config*
- Resources and execution strategies
 - errorStrategy
 - maxRetries
 - withLabel

```
process {
    // memory errors which should be retried. otherwise error out
    errorStrategy = { task.exitStatus in ((130..145) + 104) ? 'retry' : 'finish' }
    maxRetries = 3

    withLabel:process_single {
        cpus    = { 1 }
        memory  = { 8.GB * task.attempt }
        time    = { 2.h * task.attempt }
    }
    withLabel:process_low {
        cpus    = { 4 * task.attempt }
        memory  = { 8.GB * task.attempt }
        time    = { 4.h * task.attempt }
    }
    withLabel:process_medium {
        cpus    = { 8 * task.attempt }
        memory  = { 16.GB * task.attempt }
        time    = { 8.h * task.attempt }
    }
    withLabel:process_high {
        cpus    = { 12 * task.attempt }
        memory  = { 24.GB * task.attempt }
        time    = { 12.h * task.attempt }
    }
    withLabel:process_long {
        time    = { 24.h * task.attempt }
    }
    withLabel:process_high_memory {
        memory  = { 90.GB * task.attempt }
    }
    withLabel:parabricks_gpu {
        cpus    = { 18 * task.attempt }
        memory  = { 90.GB * task.attempt }
        time    = { 12.h * task.attempt }
        accelerator = 2
        maxForks = 10
    }
}
```

conf/modules.config

- Included by *nextflow.config*
- Any configuration specifically required by a module

```
process {  
  
    withName: FASTP {  
        publishDir = [  
            [  
                path: "${params.outdir}/reads_qc/fastp",  
                mode: params.publish_mode,  
            ]  
        ]  
    }  
  
    withName: BWA_MEM {  
        publishDir = [  
            [  
                path: "${params.outdir}/alignments/bwa",  
                mode: params.publish_mode,  
            ]  
        ]  
    }  
  
    withName: SCAFE {  
        containerOptions = "--bind ${params.ref_scafe_dir}:/SCAFE/resources/genome"  
        cpus = 32  
  
        publishDir = [  
            [  
                path: { "${params.outdir}/scafe" },  
                mode: "copy",  
                pattern: "output/*",  
                saveAs: {filename -> filename.replaceAll(/output\/\//, "")}  
            ]  
        ]  
    }  
}
```

The bin/ folder

- Contains executables
 - `chmod +x bin/my_script.py`
- Scripts should contain the *shebang*
 - `#!/usr/bin/env python3`
 - `#!/usr/bin/env bash`
 - `#!/usr/bin/env Rscript`
- Available in your processes

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│       └── main.nf
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The lib/ folder

- Contains groovy classes
- Automatically imported as libraries

```
// lib/Validators.groovy
class Validators {
    static boolean isValidFastq(Path file) {
        return file.name.endsWith('.fastq.gz') ||
            file.name.endsWith('.fq.gz')
    }
}
```

```
workflow {
    // Create input channel from files
    input_ch = channel.fromPath(params.input)
    .map { file ->
        // Use Validators to check if file is valid FASTQ
        if (!Validators.isValidFastq(file)) {
            log.warn "Skipping invalid FASTQ file: ${file.name}"
            return null
        }
        return file
    }
}
```

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│   │   └── main.nf
│   └── mapping/
│       └── main.nf
└── lib/
    └── Validators.groovy
├── bin/
│   └── my_script.py
├── conf/
│   ├── base.config
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The assets/ folder

- Contains small, static, accessory files
- Accessible via `${projectsDir}/assets`

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├── workflows/
│   ├── reads-qc/
│   │   └── main.nf
│   └── mapping/
│       └── main.nf
├── assets/
│   ├── my_samples.tsv
│   └── my_schema.json
├── lib/
│   └── Validators.groovy
├── bin/
│   └── my_script.py
├── conf/
│   ├── base.config
│   └── modules.config
├── main.nf
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